

Optimal-transport flow matching for unpaired PYTHIA → JEWEL jet translation

Clean calorimeter jet images, trained without pairs

October 2026

seed 0; 2 GPU hours of training; 20k held-out jets per sample; full log: FLOW_NOTES.md, results:
docs/flow/translation

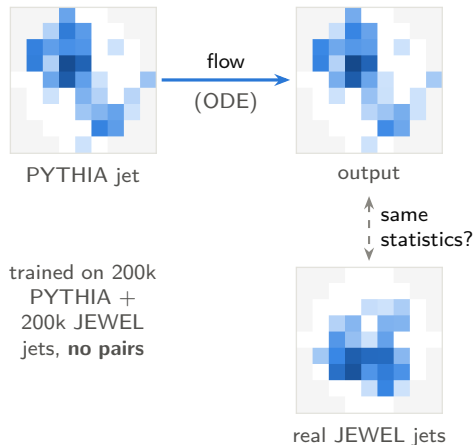
The question

Can a flow trained on **unpaired** samples map a clean PYTHIA (vacuum) jet to a JEWEL-like (medium) jet?

Two things must hold:

1. the outputs look like real JEWEL jets, statistically;
2. each output still depends on its own input jet.

There is no true JEWEL partner for a PYTHIA jet: any unpaired map is one of many with the same distributions. So this tests distributions and dependence, not a per-jet medium modification.



Method and setup

OT flow matching [1, 2]

- pair each batch of 256 PYTHIA and 256 JEWEL jets by optimal transport (least change in log energy)
- learn the velocity along straight lines between pairs (mean-squared-error loss)
- translate: solve the flow ODE (128 network calls, 6.6 ms per jet)
- network: the UVCGAN-S generator [3] with a time input, 32M parameters; 2 hours on one A6000

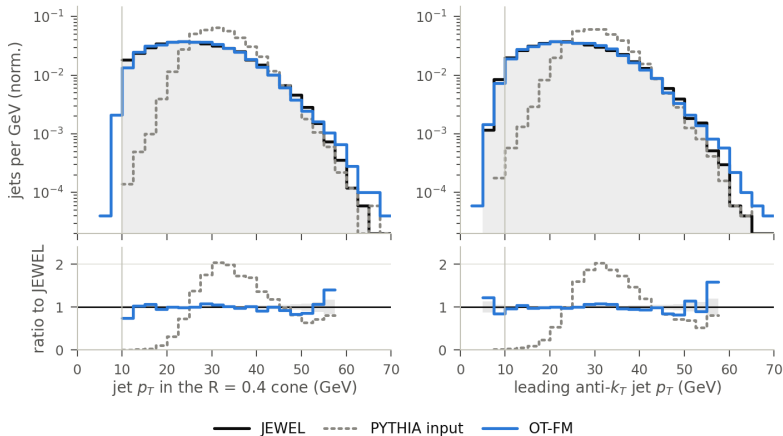
Data: clean jet images, no background

- PYTHIA 2.6M and JEWEL [4] 870k events; the leading $R = 0.4$ jet, 53 towers on a 16×16 grid
- per sample: 200k training jets, 20k held-out test jets, split by event

References for comparison

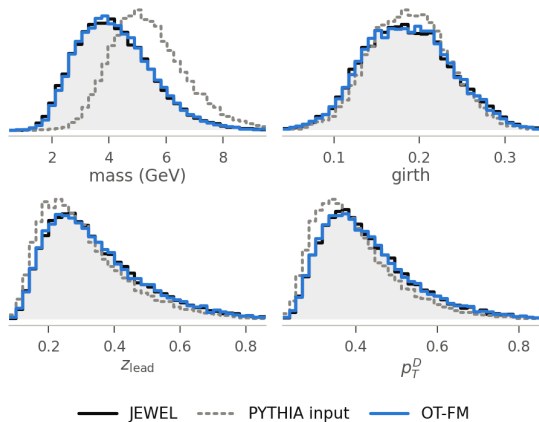
- the PYTHIA jet, unchanged
-  a random JEWEL jet: right distributions, ignores the input
- a second JEWEL sample: the best possible match

Jet p_T spectra



- the PYTHIA inputs start near 20 GeV; **OT-FM** outputs follow the JEWEL spectrum within statistical errors from 10 to about 55 GeV, for the cone and for re-found anti- k_T jets

Jet substructure

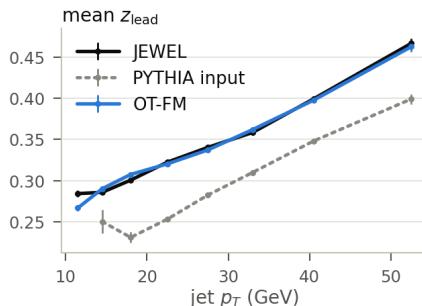
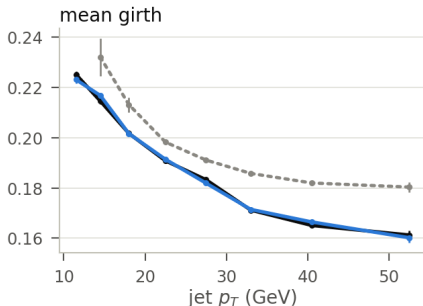


	JEWEL vs JEWEL	PYTHIA input	random JEWEL jet	OT-FM
p_T	0.011	0.55	0.019	0.025
mass	0.014	0.88	0.024	0.021
girth	0.009	0.10	0.011	0.012
z_{lead}	0.019	0.23	0.011	0.008
p_T^D	0.018	0.26	0.010	0.010
z_g	0.012	0.034	0.005	0.017
R_g	0.008	0.16	0.010	0.012

Distance to 20k held-out JEWEL jets ($W1 / \sigma$, lower is better; jets with $p_T \geq 10$ GeV). Bold: as close as a second JEWEL sample, within 3 standard deviations.

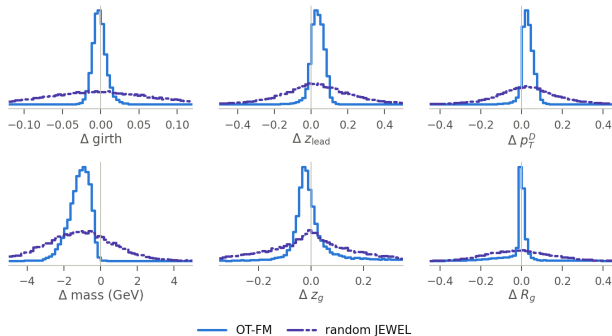
- **OT-FM** is as close to JEWEL as a second JEWEL sample in all seven observables

Jet shapes at fixed p_T



- at the same p_T , JEWEL jets are **narrower** and **harder** (higher z_{lead}) than PYTHIA jets
- **OT-FM** follows both trends in every p_T bin

How much does each jet change?

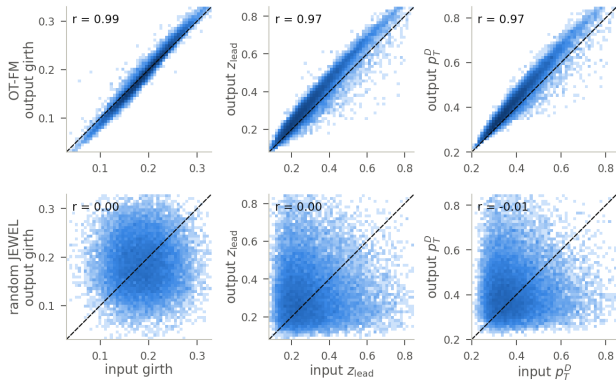


	rms of the change		ratio random / OT-FM
	— OT-FM	- - - random JEWEL jet	
girth	0.009	0.069	7.4
z_{lead}	0.051	0.203	4.0
p_T^D	0.042	0.159	3.8
mass (GeV)	1.303	2.220	1.7
z_g	0.094	0.157	1.7
R_g	0.065	0.152	2.3

Change = output minus its own input, all 20k test jets.

- both reach the JEWEL distributions, so their **average** change is the same (e.g. $z_{\text{lead}} + 0.03$, mass -1.2 GeV)
- **OT-FM** changes each jet **1.7-7 $\times$ less**: it moves a jet only as far as needed; a random JEWEL jet replaces it

Output against input, jet by jet

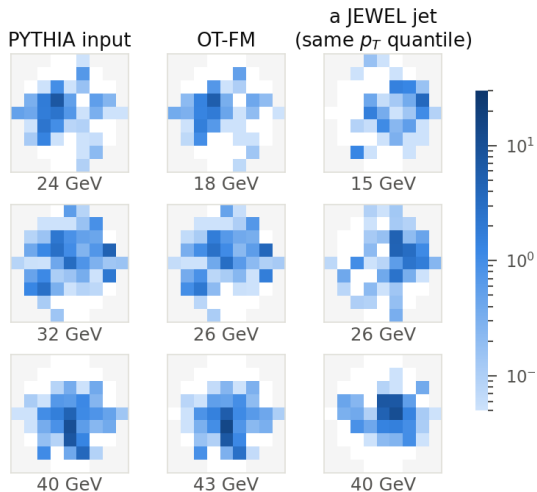


	random JEWEL jet	OT-FM
p_T	0.01	0.94
girth	0.00	0.99
mass	0.01	0.91
core fraction	0.00	0.99
leading tower	0.00	0.92
own input preferred	50%	100%

Input-output correlation, 20k test jets (core: within $\Delta R < 0.15$). Own input: share of outputs closer in shape to their own input than to another input jet.

- **OT-FM** keeps the ordering of jets and makes each slightly harder (z_{lead} , p_T^D above the diagonal)
- a random JEWEL jet has no link to its input

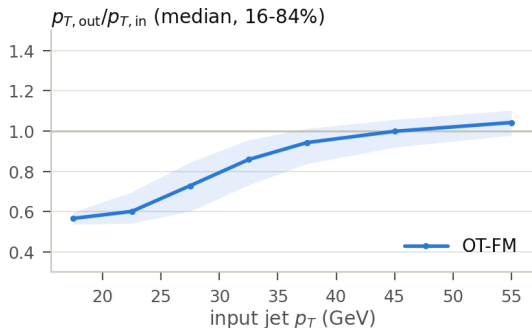
Examples



- each output keeps its input's prongs and layout
- low- p_T jets lose energy; the cores of high- p_T jets get harder

Fixed held-out inputs at p_T quantiles 0.1, 0.5 and 0.9. The JEWEL jet is a reference at the same p_T quantile, not a truth.

Caveat: the p_T change follows the two samples



- the PYTHIA sample starts near 20 GeV (presumably its generator-level jet requirement); JEWEL reaches down to 10 GeV
- **OT-FM** lowers 20-25 GeV jets by 40% and leaves jets above 40 GeV unchanged: a map between two differently selected samples, not a quenching mechanism

Summary

Shown

- trained without pairs in 2 GPU hours, **OT-FM** produces jets statistically indistinguishable from held-out JEWEL: p_T , 7 substructure observables, their p_T dependence, re-found jets
- each output stays tied to its input (correlations 0.91-0.99) and changes $1.7\text{-}7\times$ less than a random JEWEL jet would

Not shown

- a per-jet medium modification: there is no per-jet truth, and optimal transport picks the least-change map
- physics in the p_T shift: it reflects how the samples were selected
- robustness: one seed

Next: train two more seeds. Does the same PYTHIA jet get the same output, not only the same distributions?

Practical: the ODE needs 128 network calls (a 4-step solve lowers the jet p_T too much: mean 22 against 27 GeV); towers that should be empty get ~ 0.002 GeV, 0.1% of the jet energy.

References

1. Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le, *Flow Matching for Generative Modeling*, ICLR 2023, arXiv:2210.02747
2. A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, Y. Bengio, *Improving and generalizing flow-based generative models with minibatch optimal transport (OT-CFM)*, TMLR 2024, arXiv:2302.00482; code: github.com/atong01/conditional-flow-matching
3. *Stratified CycleGAN for unsupervised data decomposition (UVCGAN-S)*, arXiv:2510.23717; the generator network used here
4. K. C. Zapp, *JEWEL 2.0.0: directions for use*, Eur. Phys. J. C 74 (2014) 2762, arXiv:1311.0048